

Observation: Fare Distribution

The majority of fares are clustered between 0 and 50.

There are several outliers with fares exceeding 200, possibly 1st class passengers.

The fare distribution is right-skewed, indicating that most passengers paid lower fares

Most passengers were between 20 and 40 years old.

There is a noticeable concentration of passengers in their 20s and 30s.

Observation: Age Distribution

A few passengers were infants or elderly, showing a diverse age range.

Observation: Distribution of Categorical Features

Pclass:

Most passengers were in 3rd class, followed by 1st and then 2nd class. This indicates a larger proportion of lower-class passengers on board.

Sex:

There were significantly more male passengers than females.

The gender imbalance could affect survival analysis since survival rates differ by sex.

Embarked:

Most passengers embarked from Southampton (S), with fewer from Cherbourg (C) and Queenstown (Q).

This can reflect port popularity or passenger demographics by location.

Observation: Pairplot by Survival

Age and Fare show some separation between survivors and non-survivors. Survivors tend to have paid higher fares, which aligns with higher survival rates among

1stclass passengers.

Most of the non-survivors were from lower fare and lower class groups.

There is a wide age range in both survivor and non-survivor groups, but more survivors

appear

in the younger to middle-aged range.

No strong linear relationships are evident, but class and fare clearly have some influence on survival.

Observation: Feature Correlation

Survival is positively correlated with Fare and negatively with Pclass.

SibSp and Parch are moderately correlated — people traveling with family may have had a better survival chance

Age and Fare show weak correlation with each other.

Observation: Survival Rate by Sex

Females had a much higher survival rate than males. This supports the “women and children first” policy.

Gender is likely a key feature in predicting survival.

Observation: Survival Rate by Class

1st class passengers had the highest survival rate, followed by 2nd class, with 3rd class having the lowest.

This shows a strong correlation between class and survival — higher-class passengers had better access to lifeboats or were prioritized during evacuation.

Summary of Findings

- Survival Rate: Around 38% of passengers survived.
- Sex: Females had a significantly higher survival rate than males.
- Pclass: 1st class passengers had the highest survival rate.
- Age: Younger passengers, especially children, were more likely to survive.
- Fare: Higher fares correlated with higher survival—likely due to class/status.
- SibSp & Parch: Having family onboard slightly increased survival chances.